

Module 1 Lesson 1

Number Theory: Prime and Composite Numbers (5.1)

Objectives

In this lesson, we will learn how to

- determine divisibility;
- write the prime factorization of a composite number;
- find the greatest common divisor of two numbers;
- solve problems using the greatest common divisor;
- find the least common multiple of two numbers;
- solve problems using the least common multiple.

1. Number Theory and Natural Numbers

Number theory is primarily concerned with the properties of numbers used for counting.

The set of natural numbers is

$$\{1, 2, 3, 4, 5, \dots\}.$$

Natural numbers that are multiplied together are called factors of the resulting product.

For example,

$$4 \cdot 6 = 24.$$

So 4 and 6 are factors of 24.

Factors

If two or more numbers are multiplied to form a product, then those numbers are factors of the product.

2. Divisibility

If a and b are natural numbers, then a is divisible by b if dividing a by b leaves a remainder of 0.

This is the same as saying that b is a divisor of a , or that b divides a .
We write this as

$$b \mid a.$$

Divisibility Notation

$$b \mid a$$

means

b divides a .

In other words, $a \div b$ has remainder 0.

For example,

$$12 \mid 24$$

because $24 \div 12 = 2$ with remainder 0.

However,

$$13 \nmid 24$$

because $24 \div 13$ does not have remainder 0.

3. Rules of Divisibility

Divisibility rules help us decide quickly whether one number is divisible by another.

Divisibility Rules

Divisible By	Test
2	The last digit is 0, 2, 4, 6, or 8.
3	The sum of the digits is divisible by 3.
4	The last two digits form a number divisible by 4.
5	The number ends in 0 or 5.
6	The number is divisible by both 2 and 3.
8	The last three digits form a number divisible by 8.
9	The sum of the digits is divisible by 9.
10	The last digit is 0.
12	The number is divisible by both 3 and 4.

Example 1: Determine whether each statement is true.

(a) 5,892,796 is divisible by 2.

(b) 52,341 is divisible by 3.

(c) 3,947,136 is divisible by 4.

(d) 998,746,250 is divisible by 10.

Solution

- (a) 5,892,796 is divisible by 2.

The last digit is 6, which is even. Therefore,

True

- (b) 52,341 is divisible by 3.

Add the digits:

$$5 + 2 + 3 + 4 + 1 = 15.$$

Since 15 is divisible by 3, the original number is divisible by 3.
Therefore,

True

- (c) 3,947,136 is divisible by 4.

Look at the last two digits:

$$36.$$

Since 36 is divisible by 4, the original number is divisible by 4.
Therefore,

True

- (d) 998,746,250 is divisible by 10.

A number is divisible by 10 if the last digit is 0. Since this number ends in 0, it is divisible by 10. Therefore,

True

4. Prime and Composite Numbers

A **prime number** is a natural number greater than 1 that has only two factors: 1 and itself.

Examples of prime numbers include

2, 3, 5, 7, 11, 13.

A **composite number** is a natural number greater than 1 that is divisible by a number other than 1 and itself.

Examples of composite numbers include

4, 6, 8, 9, 10, 12.

Prime and Composite Numbers

- A prime number has exactly two positive factors: 1 and itself.
- A composite number has more than two positive factors.

5. Prime Factorization

The **prime factorization** of a number writes the number as a product of prime numbers.

Fundamental Theorem of Arithmetic

Every composite number can be expressed as a product of prime numbers in one and only one way, except for the order of the factors.

One method for finding the prime factorization of a composite number is called a **factor tree**.

Example 2: Find the prime factorization of 700.

Solution

Start by factoring 700:

$$700 = 7 \cdot 100.$$

Now factor 100:

$$100 = 10 \cdot 10.$$

Factor each 10:

$$10 = 2 \cdot 5.$$

Therefore,

$$700 = 7 \cdot 10 \cdot 10$$

$$700 = 7 \cdot 2 \cdot 5 \cdot 2 \cdot 5.$$

Rearrange the prime factors:

$$700 = 2 \cdot 2 \cdot 5 \cdot 5 \cdot 7.$$

Using exponents,

$$700 = 2^2 \cdot 5^2 \cdot 7.$$

6. Greatest Common Divisor

The greatest common divisor, or GCD, of two or more numbers is the greatest number that divides each of them.

It is also often called the greatest common factor, or GCF.

Finding the Greatest Common Divisor

To find the greatest common divisor of two or more numbers:

1. Write the prime factorization of each number.
2. Select each prime factor common to all factorizations, using the smallest exponent.
3. Multiply these selected factors.

Pairs of numbers that have 1 as their greatest common divisor are called relatively prime.

For example, the greatest common divisor of 5 and 26 is 1. Therefore, 5 and 26 are relatively prime.

7. Finding the Greatest Common Divisor

Example 3: Find the greatest common divisor of 216 and 234.

Solution

First find the prime factorization of each number.

$$216 = 2 \cdot 108$$

$$216 = 2 \cdot 2 \cdot 54$$

$$216 = 2 \cdot 2 \cdot 2 \cdot 27$$

$$216 = 2^3 \cdot 3^3.$$

Now factor 234:

$$234 = 2 \cdot 117$$

$$117 = 3 \cdot 39$$

$$39 = 3 \cdot 13.$$

Therefore,

$$234 = 2 \cdot 3^2 \cdot 13.$$

Now compare the prime factorizations:

$$216 = 2^3 \cdot 3^3$$

$$234 = 2 \cdot 3^2 \cdot 13.$$

The common prime factors are 2 and 3. Use the smallest exponents:

$$2^1 \cdot 3^2 = 2 \cdot 9 = 18.$$

Thus,

$$\boxed{\gcd(216, 234) = 18}.$$

8. Least Common Multiple

The least common multiple, or LCM, of two or more natural numbers is the smallest natural number that is divisible by all of the numbers.

Finding the Least Common Multiple

To find the least common multiple using prime factorization:

1. Write the prime factorization of each number.
2. Select every prime factor that occurs, using the greatest exponent appearing in the factorizations.
3. Multiply these selected factors.

9. Finding the Least Common Multiple

Example 4: Find the least common multiple of 144 and 300.

Solution

First find the prime factorization of each number.

$$144 = 12 \cdot 12$$

$$144 = (2^2 \cdot 3)(2^2 \cdot 3)$$

$$144 = 2^4 \cdot 3^2.$$

Now factor 300:

$$300 = 3 \cdot 100$$

$$100 = 2^2 \cdot 5^2.$$

Therefore,

$$300 = 2^2 \cdot 3 \cdot 5^2.$$

Now compare the prime factorizations:

$$144 = 2^4 \cdot 3^2$$

$$300 = 2^2 \cdot 3 \cdot 5^2.$$

For the LCM, use every prime factor with the greatest exponent:

$$\text{lcm}(144, 300) = 2^4 \cdot 3^2 \cdot 5^2.$$

Now multiply:

$$2^4 \cdot 3^2 \cdot 5^2 = 16 \cdot 9 \cdot 25.$$

$$16 \cdot 9 = 144$$

$$144 \cdot 25 = 3600.$$

Thus,

$$\boxed{\text{lcm}(144, 300) = 3600}.$$

10. Solving a Problem Using the Least Common Multiple

Example 5: A movie theater runs its films continuously. One movie runs for 80 minutes, and a second movie runs for 120 minutes. Both movies begin at 4:00 p.m. When will the movies begin again at the same time?

Solution

We need to find when both movie cycles line up again. This means we need the least common multiple of 80 and 120.

First find the prime factorization of each number:

$$80 = 8 \cdot 10$$

$$80 = 2^3 \cdot 2 \cdot 5$$

$$80 = 2^4 \cdot 5.$$

Now factor 120:

$$120 = 12 \cdot 10$$

$$120 = (2^2 \cdot 3)(2 \cdot 5)$$

$$120 = 2^3 \cdot 3 \cdot 5.$$

For the LCM, use every prime factor with the greatest exponent:

$$\text{lcm}(80, 120) = 2^4 \cdot 3 \cdot 5.$$

Now multiply:

$$2^4 \cdot 3 \cdot 5 = 16 \cdot 3 \cdot 5 = 240.$$

So the movies will begin again at the same time after 240 minutes.
Since

$$240 \text{ minutes} = 4 \text{ hours},$$

we add 4 hours to 4:00 p.m.:

$$4:00 \text{ p.m.} + 4 \text{ hours} = 8:00 \text{ p.m.}$$

Thus,

$$\boxed{8:00 \text{ p.m.}}$$

Summary

Key Ideas

- Natural numbers are $1, 2, 3, 4, 5, \dots$
- If $b \mid a$, then b divides a .
- A prime number has exactly two positive factors: 1 and itself.
- A composite number has more than two positive factors.
- Prime factorization writes a number as a product of primes.
- The greatest common divisor uses common prime factors with the smallest exponents.
- The least common multiple uses all prime factors with the greatest exponents.
- The LCM is useful for problems involving repeated cycles or events.